

Lab 2: Describing Variables in R

For this session, we will use data from the NSECE 2019 Workforce Questionnaire to learn how to describe different types of variables.

1. Set up your working environment

Load the following packages: tidyverse, ggplot2.

2. Prepare the data

Subset the NSECE data to keep the following variables:

- WF9_WORK_FT (staff full-time or part-time status)
- WF9_CHAR_GENDER (staff gender)
- WF9_WORK_MONTHS (number of months worked in the past 12 months)
- WF9_WORK_RESPECT (agreement with: “my co-workers and I are treated with respect”)
- WF9_WORK_WAGE (hourly wage)

Rename these variables to more readable names. Perform any data cleaning you find necessary. For each variable, indicate its type (nominal, ordinal, interval/ratio; discrete or continuous).

3. Describe the variables

Display each variable using the appropriate method based on the summary table discussed in class. Then answer the following questions.

For each response, report:

- Total number of observations used
 - Number of missing values (NAs)
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- a) How many staff members in the sample work full-time in center-based childcare settings?
 - b) How many staff members in the sample who identify as male work in center-based childcare settings?
 - c) What is the most common number of months worked in the past 12 months?
 - d) What percentage of staff neither agree nor disagree with the statement: “my co-workers and I are treated with respect”?
 - e) What is the shape of the distribution of hourly wages?